

## A Spindle-Exact on a gene set S

Query: 12 genes (a contiguous set) 

niche j: S meets 1 of 16 blocks



block 14: 12 of its 16 genes  
are in S (filled:  $S \times S$ )

niche j': S meets 1 of 15 blocks



block 8: 12 of its 24 genes  
are in S (filled:  $S \times S$ )

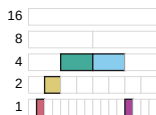
Stored per tile: block logs + block covariances.  
Per block, the  $S \times S$  part of the stored covariance  
is eigendecomposed (stored log if S covers it).

$$d_B^S(q, t) = \sqrt{\frac{\sum_b \log Q_{S \cap b} - \log T_{S \cap b} \|F\|_F^2}{\sqrt{|S|}}}$$

every tile of every niche: exact ranking

## 2 Interval index (approximate)

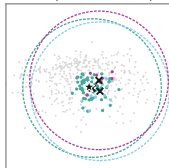
(i) index: every block (here niche j, block 14,  
16 genes)  $\rightarrow$  dyadic intervals of  $\leq 16$  genes



genes 1–16 of the block; filled: the query's pieces

per interval of s genes:  $\sim 32 \cdot \sqrt{s}$  node means;  
every tile within  $\epsilon_s$  of its node mean  
(stored: node means + one code per tile)

tiles on one 4-gene interval (PCA):  
× node mean, dashed  $\epsilon$  ball, ★ query



(ii) query  $\rightarrow$  segments (runs in a block)  
 $\rightarrow$  fewest dyadic pieces, per niche

niche j: 1 segment, 5 pieces



niche j': 3 segments, 7 pieces



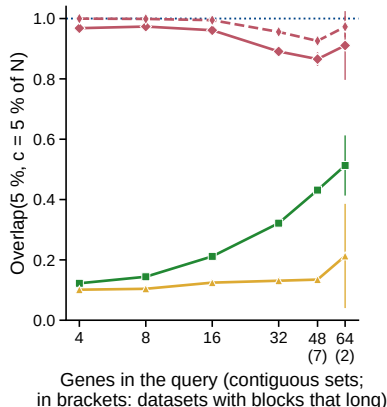
(iii) score every tile t:

$$\hat{d}(t) = \sqrt{\frac{\sum_l \log Q_l - \mu_l(t) \|F\|_F^2}{|S|}}$$

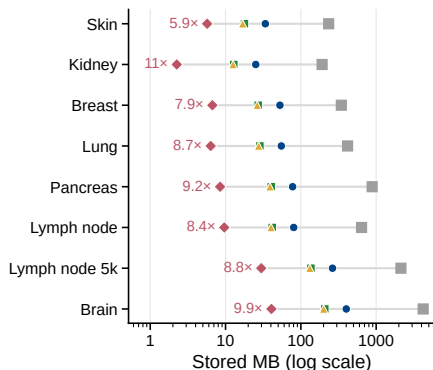
$$|\hat{d}(t) - d_{\text{own}}(t)| \leq \sqrt{\frac{\sum_l \epsilon_l^2}{|S|}} \quad (\text{here: niche j } 2.87, j' \text{ } 1.96)$$

$\mu_l(t)$ : t's node mean on piece l;  $d_{\text{own}}$ : same sum with t's own logs

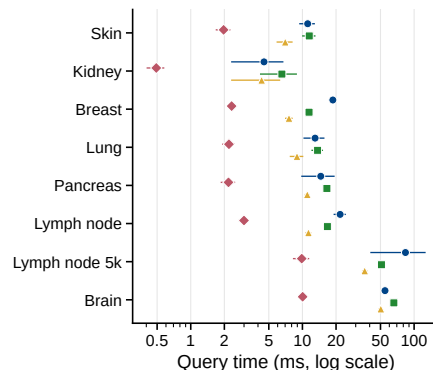
B



C



D



● Spindle-Exact (ground truth) ◆ Interval decomposition, own logs (no index) ▲ Padding + Spindle-Exact  
■ Imputation + Spindle-Exact ■ Whole covariances